

Electronics

IN ELECTRONIC SYSTEMS, INFORMATION FLOWS IN THE FORM OF PRECISELY CONTROLLED ELECTRICAL SIGNALS THROUGH CIRCUITS.

Almost all modern machines, from computers and phones to washing machines and cars, contain electronic devices of many kinds.

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Electronic components

These are components designed to handle, control, and change the amount of electric current flowing through circuits in a device. The current acts like an electrical signal, instructing the circuit and device to perform specific tasks, from adding up numbers on a calculator to displaying a word on screen. When first invented, these devices were large and bulky, and individually built and wired together. Now they have been miniaturized so that thousands can exist together on a tiny silicon microchip. When electronic circuits are designed, each component is represented by a special symbol, including the ones on the right.



△ **Diode**
This makes current flow in one direction.



△ **Connected wires**
The symbol for wires that are connected.



△ **Overlapping wires**
These wires cross but are not connected.



△ **Light-emitting diode**
This converts electrical energy to light.



△ **Amplifier**
This device increases electrical power.



△ **Transistor**
This device controls the size of current.



△ **Piezo transducer**
Converts electrical energy to sound.



△ **Fuse**
This component burns out if the circuit shorts.



△ **Thermistor**
This device converts heat to electricity.



△ **Generator**
This generates electrical voltage.



△ **AC power supply**
This supplies energy as an alternating current.



△ **DC power supply**
This supplies energy as a direct current.



△ **Inductor**
This is a type of electromagnet.



△ **Transformer**
This varies the current and voltage.



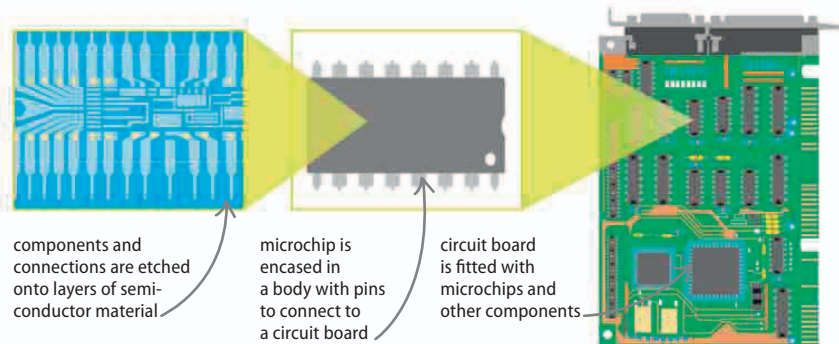
△ **Microphone**
This changes sound into electrical energy.



△ **Aerial**
This device sends or receives radio waves.

Integrated circuits

Modern electronic circuits are built onto tiny rectangles of silicon to make microchips. They are called integrated circuits because the components are all constructed together. An integrated circuit is built up from layers of different materials. Some of these layers are insulators, some are conductors, and some are semiconductors, which allow electricity to flow, but only in certain conditions. Patterns etched in the layers produce the components and their interconnections.



△ **Integrated circuit**
Electronic components are so small they are only visible under a microscope.

△ **Microchip**
This is constructed from a tiny wafer of silicon, and contains many integrated circuits.

△ **Circuit board**
Containing many microchips and other components, this forms a key part of many devices.